Does a protective skin microbiome save the population?

Building biological realism one step at a time

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The question and the method

Amphibians can be treated with skin bacteria that inhibit the chytrid fungi *Batrachochytrium dendrobatidis* (Bd) and *B. salamandrivorans* (Bsal). Would that protect a population?

The answer a model gives depends on how much biology is in it. This vignette makes that dependence visible. For each of three host–pathogen systems it:

1. starts from the simplest model of the host and the pathogen;

2. adds the biology of the host and pathogen one piece at a time, with no treatment;
3. adds the biology of the treatment one piece at a time;
4. after every step, records the same two outcomes: the population without treatment, and the population under one fixed reference treatment.

Each step adds one thing, and the step before it is recovered by switching that thing off. The final sections measure how much each addition moved the answer, and check with a factorial design whether the effect of one piece of realism depends on the others. A second vignette repeats the exercise in a single nested model, where every combination of pieces can be compared like for like.

Part	System	Simplest model
A	Bsal in fire salamanders	susceptible and infectious adults
B	Bd in common toads	Mitchell et al. (2008) pond model
C	Bd in frogs	Briggs et al. (2010) load model

> **!** Important
>
> The published models (Schmidt et al. 2017, Mitchell et al. 2008, Briggs et al. 2010) are reproduced from their papers. Everything added to them is new work in this package and is not a published result. Values taken from experiments are read from the package's parameter compendium and are named where they are used. All other added values (efficacy, coverage, spread, exposure windows, clearance) are illustrative: they show what the model does, not what a real treatment would achieve.

```
using Chytridiomycosis, Plots, Printf
using Distributions: LogNormal, quantile
using Random: Xoshiro
default(linewidth = 2, size = (720, 380), legendfontsize = 8, guidefontsize
 ↪  = 9,
    titlefontsize = 10, left_margin = 4Plots.mm, bottom_margin = 4Plots.mm)

T(; kw...) = MicrobiomeTreatment(; kw...)

# One row per step, filled in as we go. `untreated` and `treated` are the
 ↪  two outcomes;
# NaN where one does not apply.
ladder = NamedTuple[]
function step!(system, phase, added, untreated, treated, note)
    n = count(r -> r.system == system, ladder)
    push!(ladder, (; system, n, phase, added, untreated, treated, note))
    return (step = n, untreated = round(untreated, digits = 3), treated =
     ↪  round(treated, digits = 3))
end
```

```
function ladder_plot(system, xlabel)
    rows = filter(r -> r.system == system, ladder)
    k = length(rows)
    labels = ["$(r.n). $(r.added)" for r in rows]
    z(x) = isnan(x) ? 0.0 : x
    bar((1:k) .- 0.2, [z(r.untreated) for r in rows]; orientation = :h,
     ↪  bar_width = 0.38,
        label = "no treatment", color = :gray70, yticks = (1:k, labels),
 ↪  yflip = true, xlabel,
        xlims = (0, 1.05), xticks = 0:0.25:1, size = (840, 80 + 36k),
 ↪  left_margin = 6Plots.mm,
        legend = :outertop, legend_column = 2)
    bar!((1:k) .+ 0.2, [z(r.treated) for r in rows]; orientation = :h,
     ↪  bar_width = 0.38,
        label = "with treatment", color = :steelblue)
end
```

```
ladder_plot (generic function with 1 method)
```

Part A. Bsal in fire salamanders

The outcome in Part A is the mean number of adults over the ten years after Bsal arrives, as a fraction of carrying capacity. The reference treatment is: every adult treated once, on the day Bsal arrives, with bacteria that block 99% of infections for life.

Host and pathogen

Step 0. Susceptible and infectious adults

The simplest model: an adult is susceptible or infectious, infection passes by contact, and infectious animals die of the disease. This is the model of Schmidt et al. (2017) with its latent period made negligibly short.

```
p = SchmidtParameters(:schmidt2017)
p_si = Chytridiomycosis.with(p; e = 5000.0)          # latent period under
 ↪  two hours
block = T(efficacy = 0.99)

sli(q, tr; kw...) = solve(ProbioticSLIProblem(q, tr; kw...))
u0, t0 = sli(p_si, T()), sli(p_si, block; coverage = 1.0)
@printf("R0 = %.1f\n", basic_reproduction_number(p_si))
step!("Bsal", "host", "susceptible and infectious adults", u0.mean_hosts,
 ↪  t0.mean_hosts,
```

```
    "collapse; treated adults live but are not replaced")
```

```
R0 = 24.8
```

```
(step = 0, untreated = 0.003, treated = 0.338)
```

Step 1. A latent period

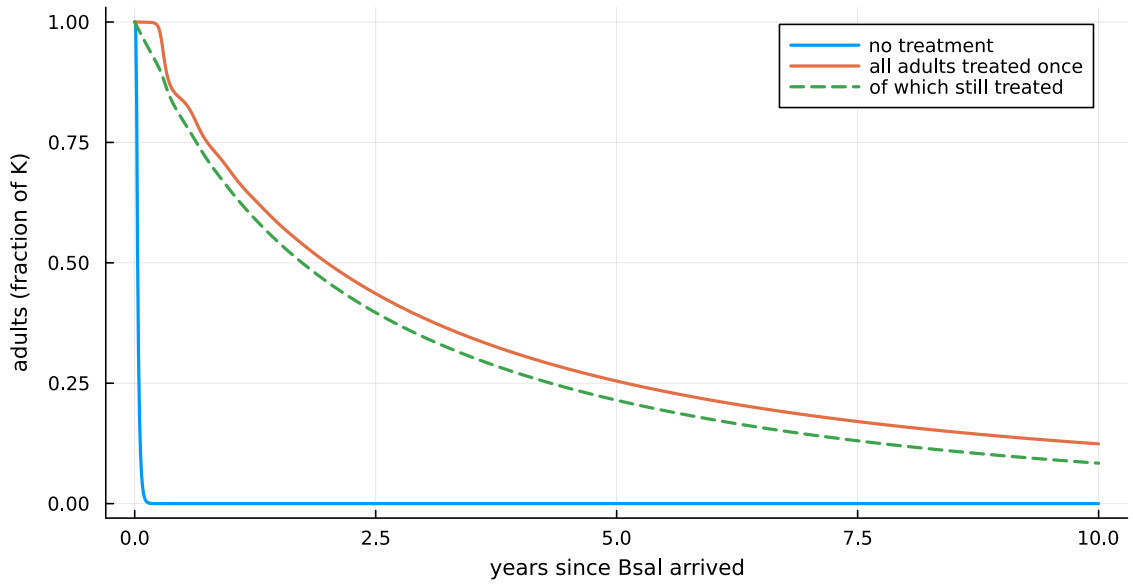
Infected animals are not infectious at once (Schmidt et al. 2017, as published).

```
u1, t1 = sli(p, T()), sli(p, block; coverage = 1.0)
@printf("R0 = %.1f\n", basic_reproduction_number(p))
step!("Bsal", "host", "+ latent period (Schmidt 2017)", u1.mean_hosts,
 ↪  t1.mean_hosts,
    "almost no change: a one-day delay against a one-week illness")
```

```
R0 = 24.8
```

```
(step = 1, untreated = 0.004, treated = 0.338)
```

```
plot(u1.t, u1.hosts ./ p.K; label = "no treatment", xlabel = "years since
 ↪  Bsal arrived",
    ylabel = "adults (fraction of K)")
plot!(t1.t, t1.hosts ./ p.K; label = "all adults treated once")
plot!(t1.t, t1.treated ./ p.K; label = "of which still treated", linestyle =
 ↪  :dash)
```



Treated adults survive, but the population still declines: recruits are born untreated and Bsal is still circulating.

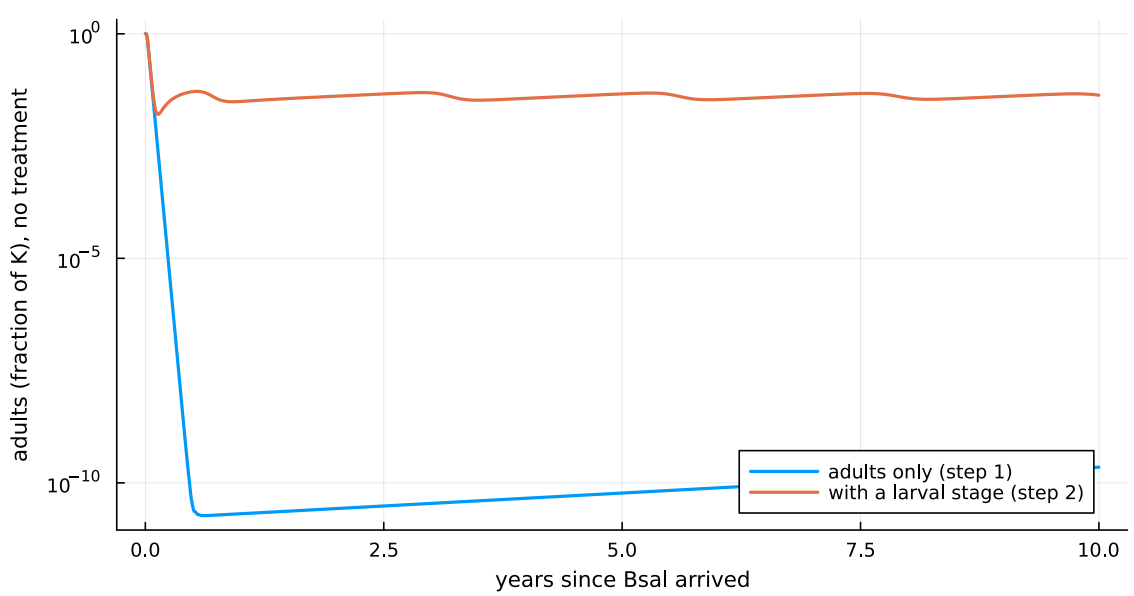
Step 2. A larval stage

Salamanders have aquatic larvae. Here they are added as a stage that Bsal does not infect.

```
clean_larvae = BsalLarvalStage(carry_over = 0.0)
lar(st, tr; kw...) = solve(ProbioticLarvalProblem(p, st, tr; kw...))
u2, t2 = lar(clean_larvae, T()), lar(clean_larvae, block; coverage = 1.0)
step!("Bsal", "host", "+ larval stage, not infected", u2.mean_adults,
 ↪  t2.mean_adults,
    "untreated population no longer vanishes: larvae are a refuge")
```

```
(step = 2, untreated = 0.044, treated = 0.336)
```

```
plot(u1.t, max.(u1.hosts ./ p.K, 1e-12); yscale = :log10, label = "adults
 ↪  only (step 1)",
    xlabel = "years since Bsal arrived", ylabel = "adults (fraction of K),
 ↪  no treatment",
    legend = :bottomright)
plot!(u2.t, u2.adults ./ p.K; label = "with a larval stage (step 2)")
```



Larvae already in the water metamorphose after the adult epidemic has burnt out. The near-extinction in the adults-only model was partly a result of having no other stage.

Step 3. Larvae can be infected

Fire salamander larvae can be infected experimentally without showing disease, and half carried the infection through metamorphosis (Wang et al. 2021; the count is read from the compendium). The transmission rate from adults to larvae is illustrative.

```
stage = BsalLarvalStage(:wang2021; adult_to_larva = 5.2)
@printf("carry-over through metamorphosis: %.1f; R0 = %.1f\n",
 ↪  stage.carry_over,
    basic_reproduction_number(p, stage))
u3, t3 = lar(stage, T()), lar(stage, block; coverage = 1.0)
step!("Bsal", "host", "+ larvae infected, carry-over 0.5", u3.mean_adults,
 ↪  t3.mean_adults,
    "small change at this carry-over")
```

```
carry-over through metamorphosis: 0.5; R0 = 24.9
```

```
(step = 3, untreated = 0.039, treated = 0.334)
```

This is the host model used for the rest of Part A.

Treatment

The reference treatment of step 3 is now made more realistic. Each step states what changed.

Step 4. Protection is lost

A treated animal reverts to untreated, with a half-life of one year.

```
a = lar(stage, T(efficacy = 0.99, half_life = 1.0); coverage = 1.0)
step!("Bsal", "treatment", "+ protection lost, half-life 1 year",
 ↪  u3.mean_adults, a.mean_adults,
    "collapse only delayed")
```

```
(step = 4, untreated = 0.039, treated = 0.133)
```

Step 5. The bacteria spread between animals

Treated animals pass the bacteria to others, so protection can reach the recruits.

```
spread = T(efficacy = 0.99, half_life = 1.0, direct = 50.0)
@printf("microbiome reproduction number %.0f, long-run coverage %.2f\n",
    microbiome_reproduction_number(spread; turnover = p.b),
     ↪  equilibrium_coverage(spread; turnover = p.b))
a = lar(stage, spread; coverage = 1.0)
step!("Bsal", "treatment", "+ bacteria spread between animals",
 ↪  u3.mean_adults, a.mean_adults,
    "full protection: the microbiome maintains itself")
```

```
microbiome reproduction number 46, long-run coverage 0.98
```

```
(step = 5, untreated = 0.039, treated = 0.999)
```

A model stopped here says: a self-spreading, strongly protective microbiome works, and nothing else does.

Step 6. What trials show: survival, not fewer infections

In challenge trials, treated animals were infected as often as untreated ones, but fewer died. The compendium holds the counts from one trial in *Rana muscosa*:

```
h = parameter_set(:harris2009ismej)
mortality_reduction(h.deaths_treated, h.n_treated, h.deaths_control,
 ↪  h.n_control; rng = Xoshiro(1))
```

```
(median = 0.9528399509303813, lower = 0.5571631729768851, upper = 0.9998974605554769)
```

So the mechanism is changed: treatment reduces disease-induced death (`tolerance`) and does not block infection. A treated, infected animal stays infectious. Loss and spread are as in step 5.

```
tolerant = T(tolerance = 0.9, half_life = 1.0, direct = 50.0)
@printf("Bsal reproduction number if everyone is treated: %.0f (untreated
 ↪  %.0f)\n",
    effective_reproduction_number(p, tolerant, 1.0),
     ↪  basic_reproduction_number(p))
a = lar(stage, tolerant; coverage = 1.0)
step!("Bsal", "treatment", "mechanism as in trials: survive, still
 ↪  infected", u3.mean_adults,
    a.mean_adults, "fails; survivors shed for longer, so transmission
 ↪  rises")
```

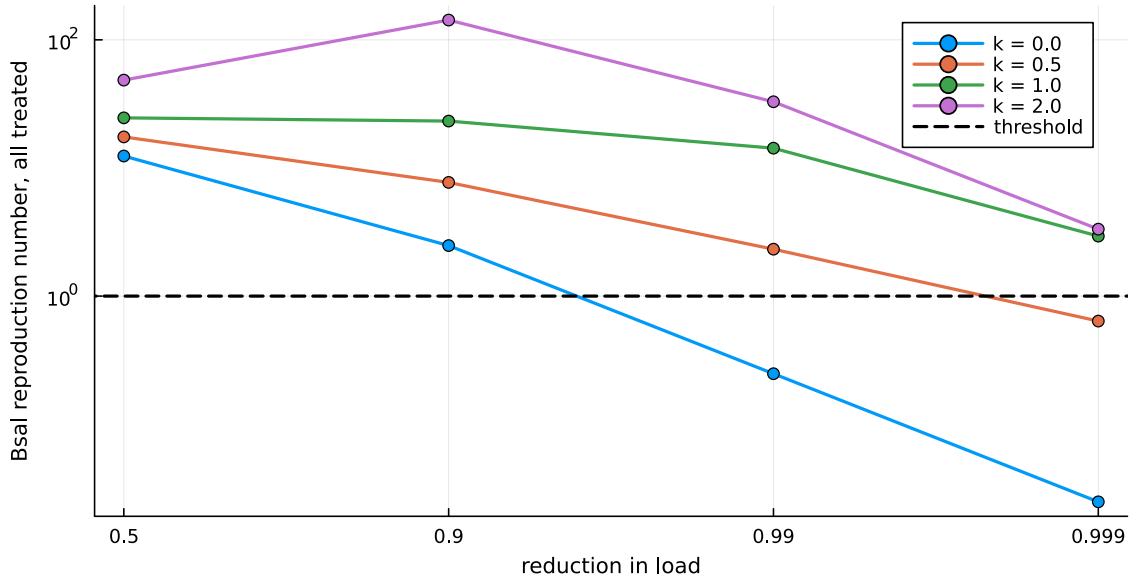
```
Bsal reproduction number if everyone is treated: 232 (untreated 25)

(step = 6, untreated = 0.039, treated = 0.025)
```

Step 7. Lower load: less death and less shedding

Bacteria that suppress the fungus should lower the load on the animal, and so reduce shedding as well as death. Both now follow from one reduction in load: shedding in proportion to load, death in proportion to load raised to a power k.

```
reductions = [0.5, 0.9, 0.99, 0.999]
ks = [0.0, 0.5, 1.0, 2.0]
R = [effective_reproduction_number(p, load_reducing_treatment(ρ;
 ↪  death_exponent = k), 1.0)
     for ρ in reductions, k in ks]
plot(string.(reductions), R; yscale = :log10, marker = :circle, xlabel =
 ↪  "reduction in load",
    ylabel = "Bsal reproduction number, all treated",
    label = permutedims(["k = $k" for k in ks]), legend = :topright)
hline!([1.0]; color = :black, linestyle = :dash, label = "threshold")
```



With death proportional to load (`k = 1`) the two effects nearly cancel: the animal sheds less but for longer.

```
a = lar(stage, load_reducing_treatment(0.99; half_life = 1.0, direct =
 ↪  50.0); coverage = 1.0)
step!("Bsal", "treatment", "+ lower load (99%): also less shedding",
 ↪  u3.mean_adults, a.mean_adults,
```

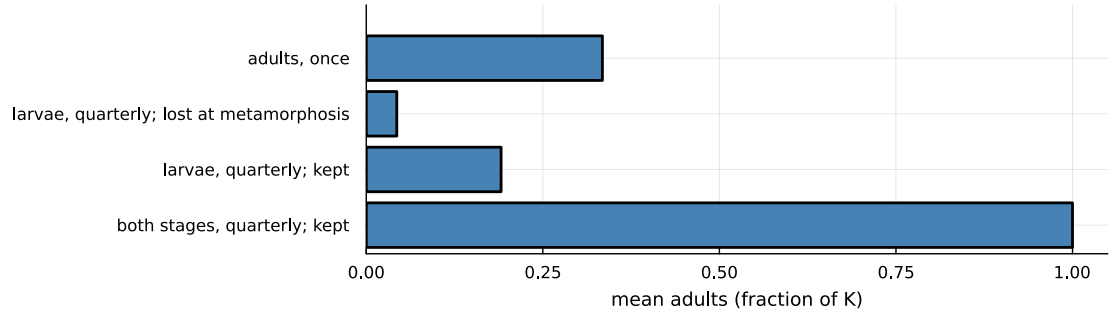
```
    "partial; survivors remain infected")
```

```
(step = 7, untreated = 0.039, treated = 0.189)
```

Step 8. Treat larvae as well

With larvae in the host model they can be treated too. Whether a treated larva keeps the bacteria through metamorphosis (`retention`) has not been measured. This step goes back to bacteria that block infection and last, with no spread, to isolate the effect of the stage.

```
quarterly = 0:0.25:9.75
options = [("adults, once", (;)),
    ("larvae, quarterly; lost at metamorphosis", (treat = :larvae,
 ↪  treat_times = quarterly)),
    ("larvae, quarterly; kept", (treat = :larvae, treat_times = quarterly,
 ↪  retention = 1.0)),
    ("both stages, quarterly; kept", (treat = :both, treat_times =
 ↪  quarterly, retention = 1.0))]
vals = [lar(stage, block; coverage = 1.0, kw...).mean_adults for (_, kw) in
 ↪  options]
bar(vals; orientation = :h, yticks = (1:4, first.(options)), yflip = true,
 ↪  legend = false,
    xlabel = "mean adults (fraction of K)", xlims = (0, 1.05), xticks =
 ↪  0:0.25:1,
    left_margin = 6Plots.mm, size = (820, 230), color = :steelblue)
```




```
step!("Bsal", "treatment", "treat both stages, kept through metamorphosis",
 ↪  u3.mean_adults, vals[4],
    "full protection with no spread at all")
```

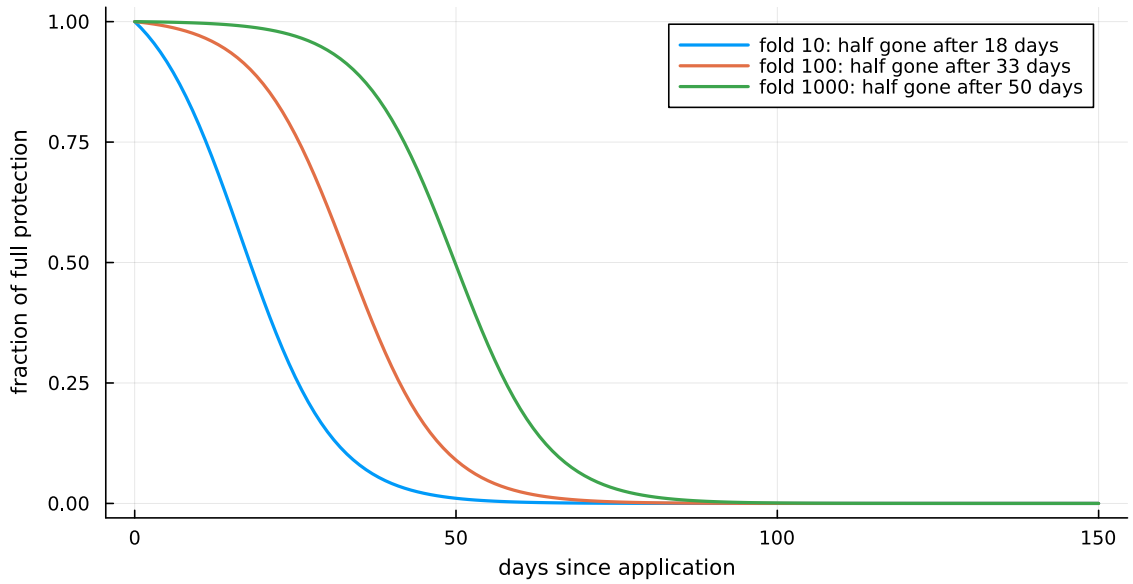
```
(step = 8, untreated = 0.039, treated = 1.0)
```

This overturns step 5. Once recruits can be treated before they become adults, the bacteria do not need to spread.

Step 9. How long the bacteria really last

Two open datasets (a field trial in *Rana sierrae*, Knapp et al. 2022, and a laboratory trial in *Xenopus laevis*, Madison et al. 2025) give the abundance of applied bacteria over time. Fitting an exponential decline gives half-lives of 3 to 14 days, and 4 to 5 days in the field. (The fits are this project's own.) Protection now wanes with abundance. `protective_fold` is how far abundance can fall before half the effect is lost; it has not been measured.

```
days = 0:1:150
plot(xlabel = "days since application", ylabel = "fraction of full
 ↪  protection", legend = :topright)
for F in (10.0, 100.0, 1000.0)
    w = T(efficacy = 0.99, abundance_half_life = 5 / 365, protective_fold =
 ↪  F)
    plot!(days, [protection(w, d / 365) for d in days];
        label = @sprintf("fold %g: half gone after %.0f days", F, 365 *
 ↪  protection_duration(w)))
end
current()
```




```
waning = T(efficacy = 0.99, abundance_half_life = 5 / 365, protective_fold =
 ↪  100.0)
a = lar(stage, waning; coverage = 1.0, treat = :both, treat_times =
 ↪  quarterly, retention = 1.0)
monthly = lar(stage, waning; coverage = 1.0, treat = :both, treat_times =
 ↪  0:(1 / 12):9.95, retention = 1.0)
@printf("dosed monthly instead of quarterly: %.2f\n", monthly.mean_adults)
```

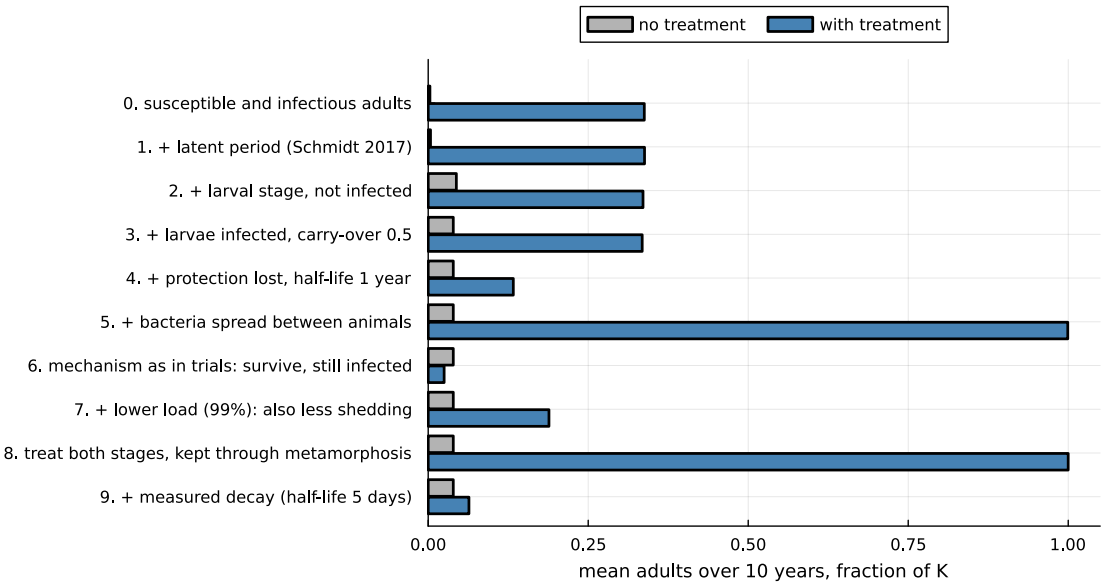
```
step!("Bsal", "treatment", "+ measured decay (half-life 5 days)",
 ↪  u3.mean_adults, a.mean_adults,
    "fails: protection lasts about a month")
```

```
dosed monthly instead of quarterly: 0.28
```

```
(step = 9, untreated = 0.039, treated = 0.064)
```

Part A summary

```
ladder_plot("Bsal", "mean adults over 10 years, fraction of K")
```



Part B. Bd in common toads

The outcome in Part B is the number of adults in years 25 to 35 after Bd arrives, as a fraction of the number without Bd. The reference treatment is: 80% of tadpoles treated at hatching every year with bacteria that block 90% of infections for the season. The parameter set is the one behind Figure 2d of Mitchell et al. (2008).

Host and pathogen

Step 0. The pond model

Mitchell's model is the simplest available for this system, though it is not as simple as step 0 of Part A: tadpoles are infected by zoospores in the pond, infected tadpoles are removed at metamorphosis, and juveniles and adults are never infected.

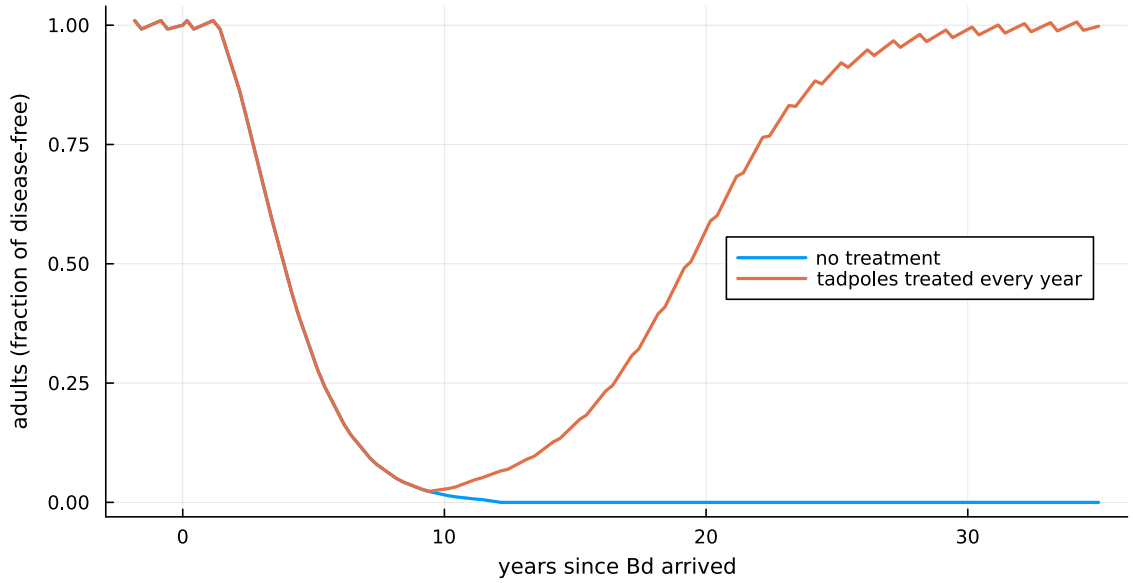
```
m = MitchellParameters(:mitchell2008; set = "fig2d")
none = PostMetamorphicInfection()
pond(tr; stages = none, kw...) = solve(ProbioticPondProblem(m, tr; stages,
 ↪  kw...))
rel(r) = r.A_mean / r.A_disease_free
ref = T(efficacy = 0.9)

u, t = pond(T()), pond(ref; coverage = 0.8)
@printf("Bd persists: untreated %s, treated %s\n", u.bd_persist,
 ↪  t.bd_persist)
step!("Bd toad", "host", "Mitchell 2008 pond model", rel(u), rel(t),
    "untreated: toads and Bd both die out; treated: toads saved, Bd gone")
```

```
Bd persists: untreated false, treated false
```

```
(step = 0, untreated = 0.0, treated = 0.975)
```

```
series(r) = (r.t[r.t .>= -2], r.A[r.t .>= -2] ./ r.A_disease_free)
plot(series(u)...; label = "no treatment", xlabel = "years since Bd
 ↪  arrived",
    ylabel = "adults (fraction of disease-free)", legend = :right)
plot!(series(t)...; label = "tadpoles treated every year")
```



Unlike Bsal, the simplest treatment works here: the infected stage is short and a new cohort can be treated each year. It does not prevent the initial crash.

Step 1. Infected metamorphs can survive

Experiments show that toadlets exposed to Bd are infected, that most die, and that some survive (Smith et al. 2022; counts from the compendium).

```
s = parameter_set(:smith2022)
survival = 1 - s.deaths_exposed / s.n_exposed
@printf("toadlet survival after exposure: %.2f\n", survival)
st1 = PostMetamorphicInfection(metamorph_survival = survival)
u, t = pond(T(); stages = st1), pond(ref; coverage = 0.8, stages = st1)
@printf("Bd persists: untreated %s, treated %s\n", u.bd_persist,
 ↪  t.bd_persist)
step!("Bd toad", "host", "+ infected metamorphs can survive (0.29)", rel(u),
 ↪  rel(t),
    "toads no longer die out, so nor does Bd; treatment now protects
     ↪  poorly")
```

```
toadlet survival after exposure: 0.29
Bd persists: untreated true, treated true
```

```
(step = 1, untreated = 0.046, treated = 0.085)
```

One added survival probability reverses both outcomes. The published “both die out” result depended on every infected animal dying; once some survive, the host persists, Bd persists with it, and blocking 90% of infections is no longer enough.

Step 2. Metamorphs are exposed as they leave the pond

```
st2 = PostMetamorphicInfection(metamorph_survival = survival,
 ↪  metamorph_exposure_days = 7.0)
u, t = pond(T(); stages = st2), pond(ref; coverage = 0.8, stages = st2)
step!("Bd toad", "host", "+ metamorphs exposed for a week on leaving",
 ↪  rel(u), rel(t),
    "little further change")
```

```
(step = 2, untreated = 0.046, treated = 0.082)
```

Step 3. Adults carry Bd between years

Breeding adults carry Bd at low prevalence in field surveys. Here adults are infected while at the pond, shed there, and clear the infection over about a year. (Treatment is not yet combined with infected adults in the package, so only the untreated outcome is shown.)

```
st3 = PostMetamorphicInfection(metamorph_survival = survival,
 ↪  metamorph_exposure_days = 7.0,
    adult_exposure_days = 30.0, adult_relative_shedding = 1.0,
 ↪  adult_clearance = 1.0)
r = solve(ToadStageProblem(m, st3))
@printf("Bd persists: %s; adult prevalence after breeding: %.2f\n",
 ↪  r.bd_persist, r.adult_prevalence)
step!("Bd toad", "host", "+ adults carry Bd between years", r.A_mean /
 ↪  r.A_disease_free, NaN,
    "untreated only; little change in this parameter set")
```

```
Bd persists: true; adult prevalence after breeding: 0.99
```

```
(step = 3, untreated = 0.044, treated = NaN)
```

The host model of step 2 is used for the rest of Part B.

Treatment

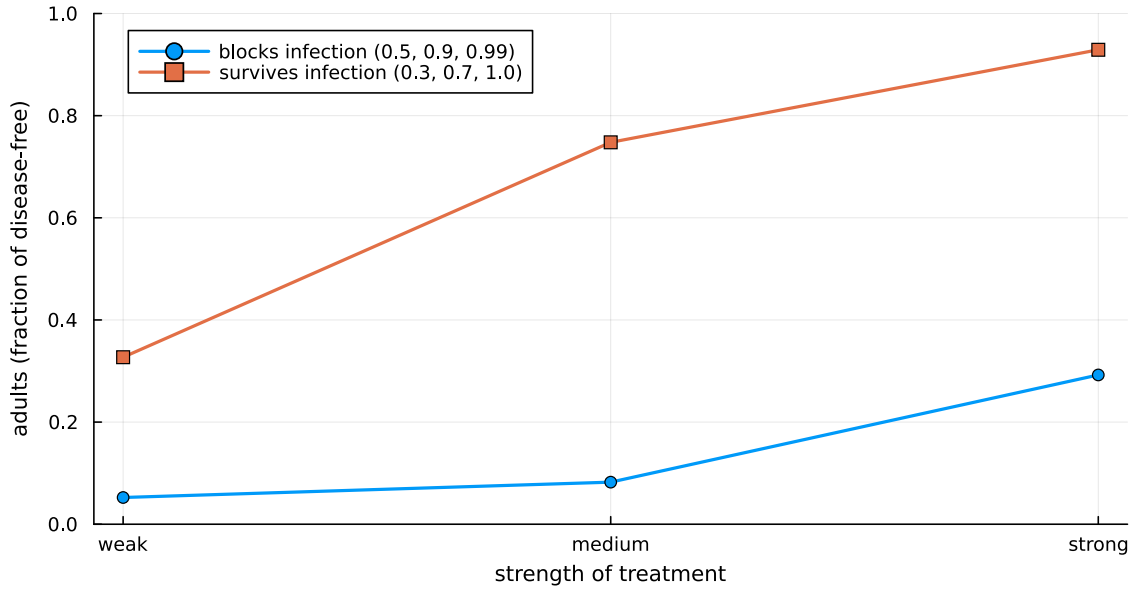
Step 4. Protection is lost

```
u = pond(T(); stages = st2)
a = pond(T(efficacy = 0.9, half_life = 0.05); coverage = 0.8, stages = st2)
step!("Bd toad", "treatment", "+ protection lost, half-life 18 days",
 ↪  rel(u), rel(a), "slightly worse")
```

```
(step = 4, untreated = 0.046, treated = 0.056)
```

Step 5. Mechanism as in trials: survival, not fewer infections

```
effs, tols = [0.5, 0.9, 0.99], [0.3, 0.7, 1.0]
blocking = [rel(pond(T(efficacy = e); coverage = 0.8, stages = st2)) for e
 ↪  in effs]
tolerating = [rel(pond(T(tolerance = τ); coverage = 0.8, stages = st2)) for
 ↪  τ in tols]
plot(["weak", "medium", "strong"], blocking; marker = :circle,
    label = "blocks infection (0.5, 0.9, 0.99)", xlabel = "strength of
 ↪  treatment",
    ylabel = "adults (fraction of disease-free)", ylims = (0, 1), legend =
 ↪  :topleft)
plot!(["weak", "medium", "strong"], tolerating; marker = :square,
    label = "survives infection (0.3, 0.7, 1.0)")
```




```
a = pond(T(tolerance = 0.7); coverage = 0.8, stages = st2)
step!("Bd toad", "treatment", "mechanism as in trials: survive, still
 ↪  infected", rel(u), rel(a),
    "much better than blocking here; Bd persists")
```

```
(step = 5, untreated = 0.046, treated = 0.748)
```

For Bsal the observed mechanism made treatment fail. Here it makes treatment work better, because Bd is going to persist anyway and what matters is how many infected animals live.

Step 6. Lower load

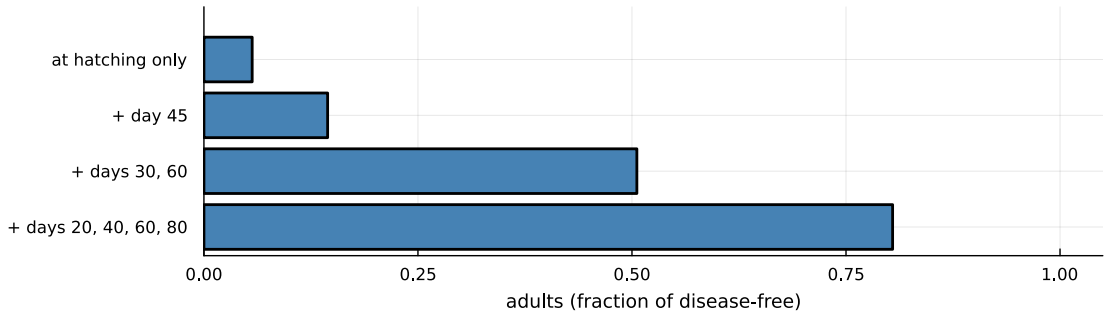
```
a = pond(load_reducing_treatment(0.7); coverage = 0.8, stages = st2)
step!("Bd toad", "treatment", "+ lower load (70%): also less shedding",
 ↪  rel(u), rel(a),
    "little changed: untreated tadpoles and the pond keep Bd going")
```

```
(step = 6, untreated = 0.046, treated = 0.751)
```

Step 7. How long the bacteria really last

With the measured decay, one dose at hatching wanes during the 95-day tadpole season.

```
waning_t = T(tolerance = 0.7, abundance_half_life = 5 / 365, protective_fold
 ↪  = 100.0)
schedules = [("at hatching only", Float64[]), ("+ day 45", [45.0]), ("+ days
 ↪  30, 60", [30.0, 60.0]),
    ("+ days 20, 40, 60, 80", [20.0, 40.0, 60.0, 80.0])]
doses = [rel(pond(waning_t; coverage = 0.8, stages = st2, redose_days = d))
 ↪  for (_, d) in schedules]
bar(doses; orientation = :h, yticks = (1:4, first.(schedules)), yflip =
 ↪  true, legend = false,
    xlabel = "adults (fraction of disease-free)", xlims = (0, 1.05), xticks
 ↪  = 0:0.25:1,
    left_margin = 6Plots.mm, size = (820, 230), color = :steelblue)
```




```
step!("Bd toad", "treatment", "+ measured decay, one dose at hatching",
 ↪  rel(u), doses[1],
    "protection gone before the season ends")
```

```
(step = 7, untreated = 0.046, treated = 0.056)
```

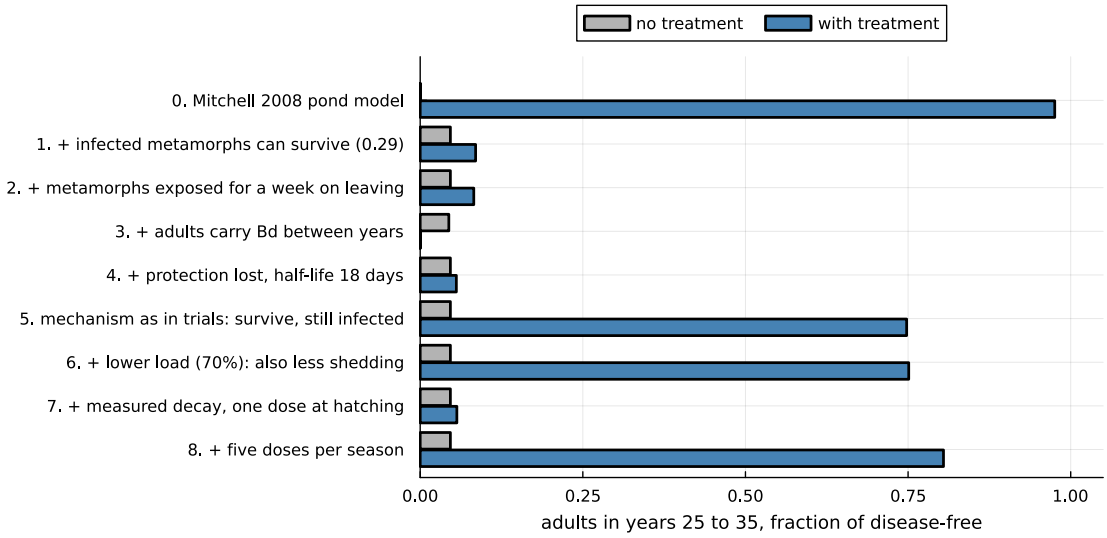
Step 8. Several doses in a season

```
step!("Bd toad", "treatment", "+ five doses per season", rel(u), doses[4],
 ↪  "most of the benefit restored")
```

```
(step = 8, untreated = 0.046, treated = 0.804)
```

Part B summary

```
ladder_plot("Bd toad", "adults in years 25 to 35, fraction of disease-free")
```



Part C. Bd in frogs

The outcome in Part C is the fraction of 80 frogs sharing one pool that are alive after two years. Disease here depends on the fungal load on each frog. The reference treatment is introduced at step 2.

Host and pathogen

Step 0. Load on each frog

In the model of Briggs et al. (2010) each frog carries a load of sporangia. Sporangia release zoospores into a shared pool, zoospores infect frogs, and a frog dies when its load reaches a lethal level. Loads run away when there are enough frogs to keep the pool topped up.

```
pb = BriggsParameters(:briggs2010)
N, hsat, ε = 80, 500.0, 1000.0
tipping(c; n = N) = solve(LoadTippingProblem(pb; c, h = hsat, N = n, ε,
 ↪  tspan = (0.0, 730.0)))
alive(r) = count(r.alive) / length(r.alive)

@printf("frogs needed for loads to run away: %.0f\n",
 ↪  host_density_threshold(pb))
r0 = tipping(0.0)
step!("Bd frog", "host", "Briggs 2010 load model", alive(r0), NaN, "all die:
 ↪  80 frogs is above the threshold of 23")
```

```
frogs needed for loads to run away: 23
```

```
(step = 0, untreated = 0.0, treated = NaN)
```

Step 1. Frogs clear some of their load

Frogs have defences. Here each frog removes sporangia at a rate that saturates as load rises: defences cope with a light infection and are overwhelmed by a heavy one (this package's load tipping model; the clearance values are illustrative).

```
c = 100.0
@printf("frogs needed for loads to run away: %d\n", critical_host_number(pb;
 ↪  c, h = hsat, ε))
r1 = tipping(c)
r1small = tipping(c; n = 50)
@printf("alive after two years: %.2f of 80 frogs, %.2f of 50 frogs\n",
 ↪  alive(r1), alive(r1small))
step!("Bd frog", "host", "+ saturating clearance by the frog", alive(r1),
 ↪  NaN,
    "threshold rises from 23 to 68 frogs; 80 still all die")
```

```
frogs needed for loads to run away: 68
alive after two years: 0.00 of 80 frogs, 1.00 of 50 frogs
```

```
(step = 1, untreated = 0.0, treated = NaN)
```

Step 2. Frogs differ

Frogs now differ in how strong their defences are (lognormal, same median). The reference treatment for Part C starts here: 40% of frogs carry bacteria that triple their clearance, for life.

```
cs = quantile.(LogNormal(log(c), 0.5), ((1:N) .- 0.5) ./ N)
r2 = tipping(cs)
frogs(tr; kw...) = solve(ProbioticLoadProblem(pb, tr; c, c_treated = 3c, h
 ↪  = hsat, N, ε, days = 730, kw...);
    rng = Xoshiro(1))
ref_c = frogs(T(); coverage = 0.4)
step!("Bd frog", "host", "+ frogs differ in their defences", alive(r2), NaN,
    "a few of the best-defended survive")
```

```
(step = 2, untreated = 0.062, treated = NaN)
```

With identical frogs a die-off is all or nothing; with differences it is partial, and it stops when too few frogs are left to sustain the pool. The treatment steps use identical frogs (step 1), because treatment and host differences are not yet combined in the package.

Treatment

Step 3. A treated class

```
step!("Bd frog", "treatment", "treated class: 40% of frogs, lasts for life",
 ↪  alive(r1),
    ref_c.alive[end] / N, "all survive, including the untreated: protection
 ↪  is shared")
```

```
(step = 3, untreated = 0.0, treated = 1.0)
```

Because disease depends on the shared pool, treating 40% protects everyone.

Step 4. Protection is lost

```
a = frogs(T(half_life = 30.0); coverage = 1.0)
@printf("first death: day %d (untreated: day %d)\n", minimum(filter(>(0),
 ↪  a.death_day)),
    ceil(Int, minimum(r1.death_time)))
step!("Bd frog", "treatment", "+ protection lost, half-life 30 days",
 ↪  alive(r1), a.alive[end] / N,
    "die-off delayed by about three months, not prevented")
```

```
first death: day 241 (untreated: day 156)
```

```
(step = 4, untreated = 0.0, treated = 0.0)
```

Step 5. Bacteria as a state of each frog

Until now every treated animal has been simply “treated”. Each frog now carries its own abundance of the applied bacteria. It declines at the measured rate (half-life 5 days), and frogs exchange bacteria through the pond. Exchange moves bacteria between frogs without creating any, so they spread but do not sustain themselves. That is what the field trial showed: eleven days after most frogs in a pond were bathed, frogs that were never bathed carried about 0.63 times as much of the bacterium as bathed ones, and it then declined in both.

```
baths(; x = 0.1, F = 100.0, coverage = 0.75, dose_days = [0], days = 730) =
    solve(ProbioticAbundanceProblem(pb, T(abundance_half_life = 5.0,
     ↪  protective_fold = F, direct = x);
        c, c_treated = 3c, h = hsat, N, ε, coverage, dose_days, days))
for x in (0.0, 0.1, 1.0)
    r = baths(; x, days = 30)
    @printf("exchange %.1f per day: unbathed/bathed abundance on day 11 =
     ↪  %.2f\n", x,
        r.abundance_unbathed[12] / r.abundance_bathed[12])
end
```

```
exchange 0.0 per day: unbathed/bathed abundance on day 11 = 0.00
exchange 0.1 per day: unbathed/bathed abundance on day 11 = 0.60
exchange 1.0 per day: unbathed/bathed abundance on day 11 = 1.00
```

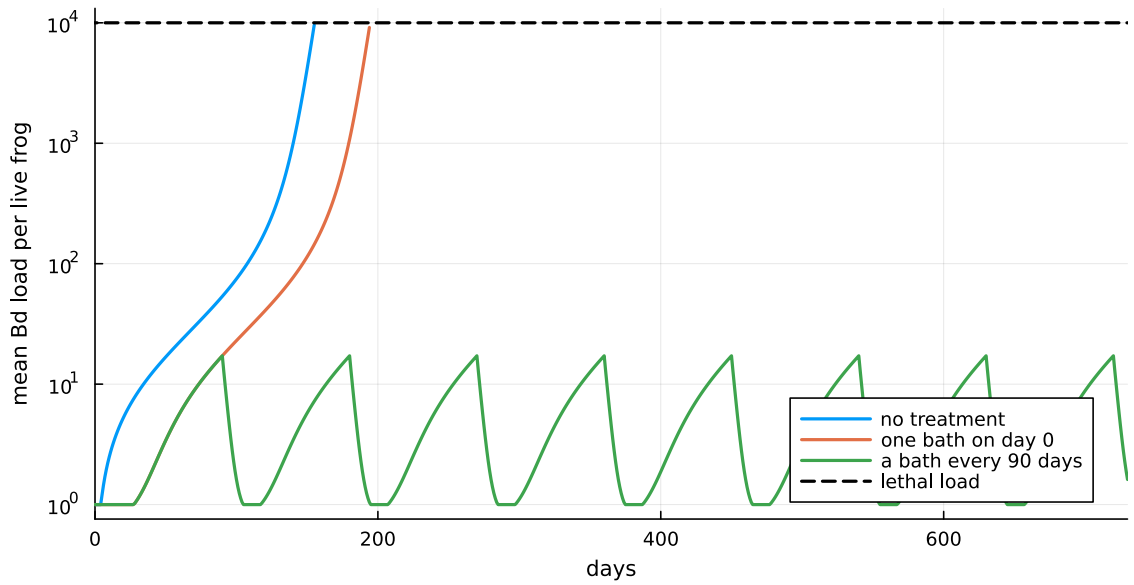
An exchange rate of about 0.1 per day reproduces the field ratio and is used below.

```
one = baths()
step!("Bd frog", "treatment", "+ bacteria on each frog, measured decay; one
 ↪  bath", alive(r1),
    one.alive[end] / N, "die-off delayed by about six weeks")
```

```
(step = 5, untreated = 0.0, treated = 0.0)
```

Step 6. Repeated baths

```
untreated, every90 = baths(coverage = 0.0), baths(dose_days = 0:90:729)
plot(untreated.t, max.(untreated.mean_load, 1.0); yscale = :log10, label =
 ↪  "no treatment",
    xlabel = "days", ylabel = "mean Bd load per live frog", legend =
 ↪  :bottomright, xlims = (0, 730))
plot!(one.t, max.(one.mean_load, 1.0); label = "one bath on day 0")
plot!(every90.t, max.(every90.mean_load, 1.0); label = "a bath every 90
 ↪  days")
hline!([pb.Smax]; color = :black, linestyle = :dash, label = "lethal load")
```




```
first_death(r) = any(>(0), r.death_day) ? minimum(filter(>(0), r.death_day))
 ↪  : 0
println("| Baths | Alive after 2 years (of 80) | First death (day) |")
println("|---|---|---|")
for (label, r) in (("none", untreated), ("one, day 0", one), ("every 240
 ↪  days", baths(dose_days = 0:240:729)),
        ("every 150 days", baths(dose_days = 0:150:729)), ("every 90 days",
 ↪  every90),
        ("every 90 days, 10% of frogs", baths(coverage = 0.1, dose_days =
 ↪  0:90:729)),
        ("every 90 days, 10% of frogs, no exchange", baths(coverage = 0.1, x
 ↪  = 0.0, dose_days = 0:90:729)),
        ("one, day 150 (loads already high)", baths(dose_days = [150])))
    @printf("| %s | %d | %s |\n", label, r.alive[end], first_death(r) == 0 ?
     ↪  "none" : string(first_death(r)))
end
```

Baths	Alive after 2 years (of 80)	First death (day)
none	0	156
one, day 0	0	195
every 240 days	0	195
every 150 days	80	none
every 90 days	80	none
every 90 days, 10% of frogs	80	none
every 90 days, 10% of frogs, no exchange	8	184
one, day 150 (loads already high)	0	157

```
step!("Bd frog", "treatment", "+ a bath every 90 days", alive(r1),
 ↪  every90.alive[end] / N,
    "all survive: each bath resets the slow build-up of load")
```

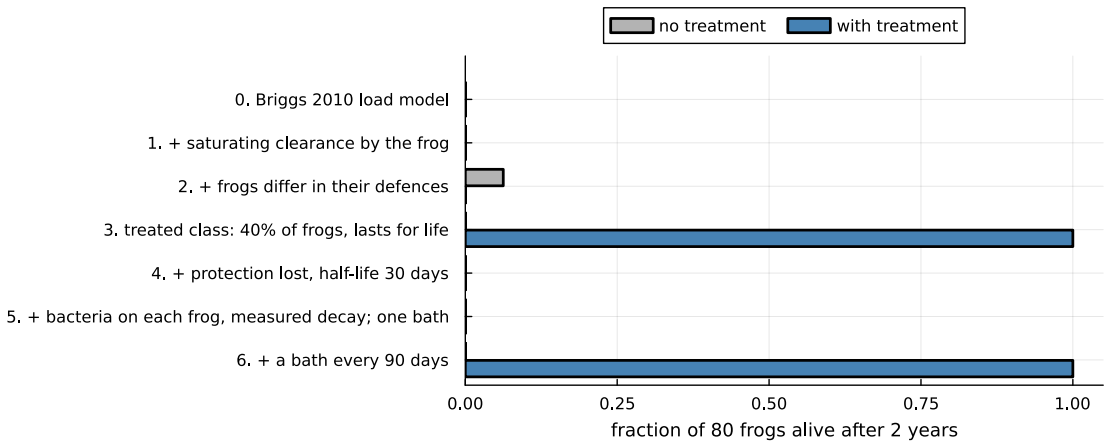
```
(step = 6, untreated = 0.0, treated = 1.0)
```

This is the one system where bacteria that last only weeks could still work. Load takes months to build; a short burst of extra clearance knocks it back, and the build-up starts again. Because frogs exchange bacteria, bathing a small part of the population protects the rest.

Two cautions. The slow build-up is for a population that starts clean. In the field trial the pond was in the middle of an outbreak, treated frogs were surrounded by heavily infected ones, and their loads were back within two months; the last row of the table is the model's version of treating too late. And the size of the clearance effect of the bacteria is assumed, not measured.

Part C summary

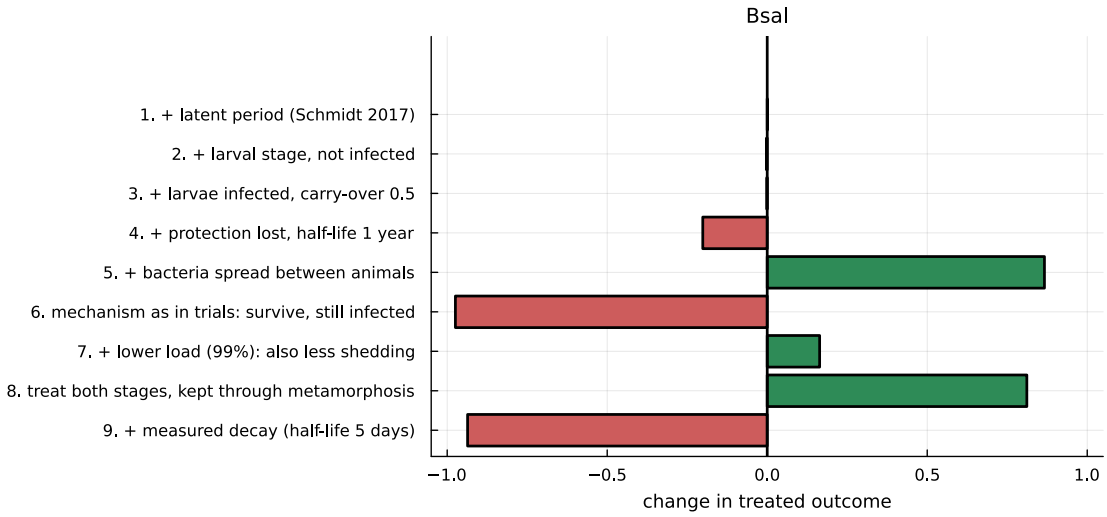
```
ladder_plot("Bd frog", "fraction of 80 frogs alive after 2 years")
```



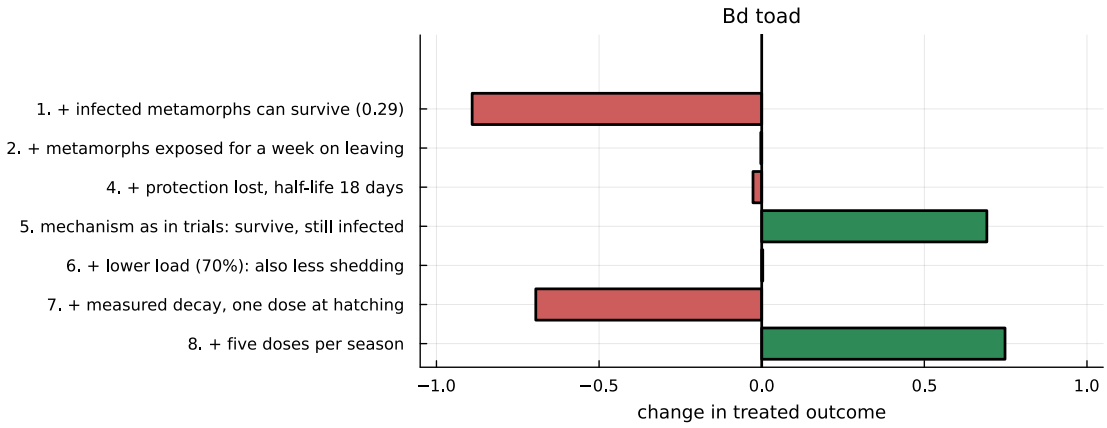
How much did each step move the answer?

The change in the treated outcome at each step, for each system. Bars to the right are steps that made treatment look better; bars to the left, worse.

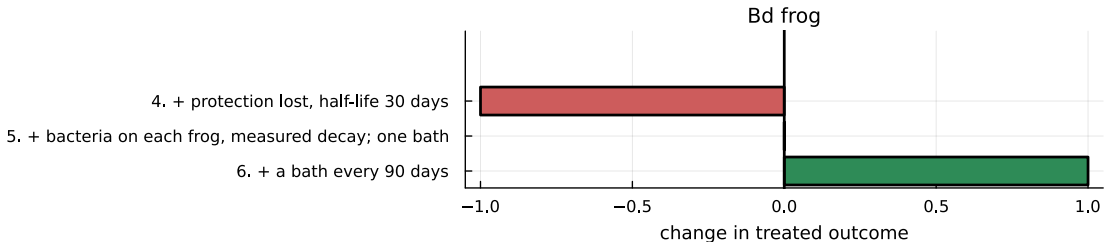
```
function change_plot(system)
    rows = filter(r -> r.system == system && !isnan(r.treated), ladder)
    d = [rows[i].treated - rows[i - 1].treated for i in 2:length(rows)]
    labels = ["$(rows[i].n). $(rows[i].added)" for i in 2:length(rows)]
    bar(d; orientation = :h, yticks = (1:length(d), labels), yflip = true,
     ↪  legend = false,
        xlims = (-1.05, 1.05), xlabel = "change in treated outcome", title
 ↪  = system,
        color = [x >= 0 ? :seagreen : :indianred for x in d], left_margin =
 ↪  6Plots.mm,
        size = (840, 70 + 34 * length(d)))
    vline!([0.0]; color = :black)
end
change_plot("Bsal")
```




```
change_plot("Bd toad")
```




```
change_plot("Bd frog")
```



Large bars point both ways in every system. No single addition dominates, and the sign of a step depends on the system: the observed mechanism (survive, still infected) is the largest negative step for Bsal and a large positive step for toads.

Does the effect of one piece of realism depend on the others?

A ladder adds things in one order. To check that the order is not driving the conclusions, four pieces of treatment realism are switched on and off in every combination for Bsal, on the full host model of step 3:

- mechanism: blocks infection (0.99), or survive but stay infected (0.99);
- duration: lasts for life, or measured decay (half-life 5 days);
- spread: none, or between animals at 50 per year;
- stages: adults once, or both stages every quarter with the bacteria kept through metamorphosis.

```
levels = (
    mechanism = ["blocks", "survive"],
    duration = ["lasting", "decays"],
    spread = ["none", "spreads"],
    stages = ["adults", "both"],
)
```

```
function factorial_run(mech, dur, spr, stg)
    tr = T(; (mech == "blocks" ? (efficacy = 0.99,) : (tolerance =
 ↪  0.99,))...,
        (dur == "decays" ? (abundance_half_life = 5 / 365, protective_fold =
 ↪  100.0) : (;))...,
        direct = spr == "spreads" ? 50.0 : 0.0)
    kw = stg == "both" ? (treat = :both, treat_times = quarterly, retention
 ↪  = 1.0) : (;)
    return lar(stage, tr; coverage = 1.0, kw...).mean_adults
end
design = [(; mech, dur, spr, stg, outcome = factorial_run(mech, dur, spr,
 ↪  stg))
          for mech in levels.mechanism, dur in levels.duration, spr in
 ↪  levels.spread, stg in levels.stages]
design = sort(vec(design); by = r -> -r.outcome)
nothing
```

```
println("| Mechanism | Duration | Spread | Stages treated | Mean adults |")
println("|---|---|---|---|---|")
for r in design
    @printf("| %s | %s | %s | %s | %.2f |\n", r.mech, r.dur, r.spr, r.stg,
     ↪  r.outcome)
end
```

Mechanism	Duration	Spread	Stages treated	Mean adults
blocks	lasting	spreads	both	1.00
blocks	lasting	none	both	1.00
blocks	lasting	spreads	adults	1.00
blocks	lasting	none	adults	0.33
survive	lasting	spreads	both	0.27
survive	lasting	none	both	0.22
survive	lasting	spreads	adults	0.20
survive	lasting	none	adults	0.13
blocks	decays	spreads	both	0.07
blocks	decays	none	both	0.06
blocks	decays	spreads	adults	0.05
blocks	decays	none	adults	0.05
survive	decays	spreads	adults	0.04
survive	decays	none	adults	0.04
survive	decays	none	both	0.03
survive	decays	spreads	both	0.03

The average effect of switching each piece on, and how much that effect varies with the other three:

```
function effect(field, on)
    diffs = Float64[]
    for r in design
        getfield(r, field) == on || continue
        partner = only(filter(q -> all(getfield(q, f) == getfield(r, f) for
 ↪  f in (:mech, :dur, :spr, :stg) if f != field) &&
            getfield(q, field) != on, design))
        push!(diffs, r.outcome - partner.outcome)
    end
    return diffs
end
println("| Switched on | Mean change | Smallest | Largest |")
println("|---|---|---|---|")
for (field, on, label) in ((:mech, "survive", "survive instead of block"),
 ↪  (:dur, "decays", "measured decay"),
        (:spr, "spreads", "spread between animals"), (:stg, "both", "treat
 ↪  both stages"))
    d = effect(field, on)
    @printf("| %s | %+.2f | %+.2f | %+.2f |\n", label, sum(d) / length(d),
     ↪  minimum(d), maximum(d))
end
```

Switched on	Mean change	Smallest	Largest
survive instead of block	-0.32	-0.80	-0.00
measured decay	-0.47	-0.95	-0.09
spread between animals	+0.10	-0.00	+0.67
treat both stages	+0.10	-0.01	+0.67

The ranges are wide: the effect of each piece depends strongly on the others. Spread, for example, matters a great deal for a lasting, infection-blocking treatment given to adults, and hardly at all once the bacteria decay in days. (In the model, animals that acquire bacteria from others share the dosing clock of the treated ones, so spread cannot outlast the decay.) A piece of realism therefore cannot be judged by adding it to a simple model alone.

All steps

```
fmt(x) = isnan(x) ? "" : @sprintf("%.2f", x)
println("| System | Step | Phase | Added | No treatment | With treatment |
 ↪  What changed |")
println("|---|---|---|---|---|---|---|")
for r in ladder
```

```
    println("| ", join((r.system, r.n, r.phase, r.added, fmt(r.untreated),
    ↪  fmt(r.treated), r.note), " | "), " |")
end
```

System	Step	Phase	Added	No treatment	With treatment	What changed
Bsal	0	host	susceptible and infectious adults	0.00	0.34	collapse; treated adults live but are not replaced
Bsal	1	host	+ latent period (Schmidt 2017)	0.00	0.34	almost no change: a one-day delay against a one-week illness
Bsal	2	host	+ larval stage, not infected	0.04	0.34	untreated population no longer vanishes: larvae are a refuge
Bsal	3	host	+ larvae infected, carry-over 0.5	0.04	0.33	small change at this carry-over
Bsal	4	treatment	+ protection lost, half-life 1 year	0.04	0.13	collapse only delayed
Bsal	5	treatment	+ bacteria spread between animals	0.04	1.00	full protection: the micro-biome maintains itself

System	Step	Phase	Added	No treatment	With treatment	What changed
Bsal	6	treatment	mechanism as in trials: survive, still infected	0.04	0.02	fails; survivors shed for longer, so transmission rises
Bsal	7	treatment	+ lower load (99%): also less shedding	0.04	0.19	partial; survivors remain infected
Bsal	8	treatment	treat both stages, kept through metamorphosis	0.04	1.00	full protection with no spread at all
Bsal	9	treatment	+ measured decay (half-life 5 days)	0.04	0.06	fails: protection lasts about a month
Bd toad	0	host	Mitchell 2008 pond model	0.00	0.98	untreated: toads and Bd both die out; treated: toads saved, Bd gone
Bd toad	1	host	+ infected metamorphs can survive (0.29)	0.05	0.09	toads no longer die out, so nor does Bd; treatment now protects poorly
Bd toad	2	host	+ metamorphs exposed for a week on leaving	0.05	0.08	little further change

System	Step	Phase	Added	No treatment	With treatment	What changed
Bd toad	3	host	+ adults carry Bd between years	0.04		untreated only; little change in this parameter set
Bd toad	4	treatment	+ protection lost, half-life 18 days	0.05	0.06	slightly worse
Bd toad	5	treatment	mecha- nism as in trials: survive, still infected	0.05	0.75	much better than blocking here; Bd persists
Bd toad	6	treatment	+ lower load (70%): also less shedding	0.05	0.75	little changed: untreated tadpoles and the pond keep Bd going
Bd toad	7	treatment	+ measured decay, one dose at hatching	0.05	0.06	protection gone before the season ends
Bd toad	8	treatment	+ five doses per season	0.05	0.80	most of the benefit restored
Bd frog	0	host	Briggs 2010 load model	0.00		all die: 80 frogs is above the threshold of 23
Bd frog	1	host	+ saturating clearance by the frog	0.00		threshold rises from 23 to 68 frogs; 80 still all die

System	Step	Phase	Added	No treatment	With treatment	What changed
Bd frog	2	host	+ frogs differ in their defences	0.06		a few of the best-defended survive
Bd frog	3	treatment	treated class: 40% of frogs, lasts for life	0.00	1.00	all survive, including the untreated: protection is shared
Bd frog	4	treatment	+ protection lost, half-life 30 days	0.00	0.00	die-off delayed by about three months, not prevented
Bd frog	5	treatment	+ bacteria on each frog, measured decay; one bath	0.00	0.00	die-off delayed by about six weeks
Bd frog	6	treatment	+ a bath every 90 days	0.00	1.00	all survive: each bath resets the slow build-up of load

What the exercise shows

1. The answer changed at almost every step, and reversed more than once. For Bsal, a model stopped at step 5 says a self-spreading microbiome is essential; at step 8, that spread is unnecessary; at step 9, that neither is achievable with bacteria that last a few weeks.
2. Host biology can move the answer as much as treatment biology, but not everywhere. A larval stage changed the untreated fate of the salamanders. One survival probability for infected toad metamorphs reversed both the untreated outcome and the value of the reference treatment. The second vignette tests this like for like in a single model and finds that it depends on the regime: for Bsal at the published host density, six pieces of host biology barely change the outcome, and near the invasion threshold they change it a great deal. Its stochastic version also shows that chance can matter more than any of them: with

whole animals, blocking infection in every adult on arrival nearly always eliminates Bsal, where the equations of step 1 here leave about a third of the population.
3. How protection works matters more than how strong it is, and its effect has no fixed sign. Letting animals survive infection is the worst change for Bsal and one of the best for toads.
4. Effects interact. The factorial shows that the value of spread, of treating larvae or of a lasting treatment each depends on the other pieces.
5. The systems differ for structural reasons. A toad cohort passes through a short infected stage once a year and can be re-treated. A salamander must be protected for life against a pathogen that kills in a week. In frogs, where disease follows a load that builds slowly, short-lived bacteria can still work if they are re-applied.

What is not in the models

- In Parts A and B bacterial abundance is a clock since the last dose, not a state of each animal. Only the frog model of Part C tracks it per animal.
- The fold-drop in abundance at which protection is lost has not been measured, nor has the effect of the bacteria on clearance, nor whether applied bacteria survive metamorphosis.
- The Bsal model has no recovery from infection and no years between metamorphosis and maturity. The toad model has thresholds standing in for extinction, and infected adult toads are not yet combined with treatment. The frog treatment steps use identical frogs.
- The three systems do not share one model structure, so Part B starts from a more complex model than Part A. The second vignette uses one nested model, for the salamander system, in which every piece of biology is a switch.
- No fade-out rule is applied in Part A, so densities can fall to meaningless values and recover. The second vignette shows what such a rule changes.
- Everything is deterministic except the choice of which frogs are treated in Part C.

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